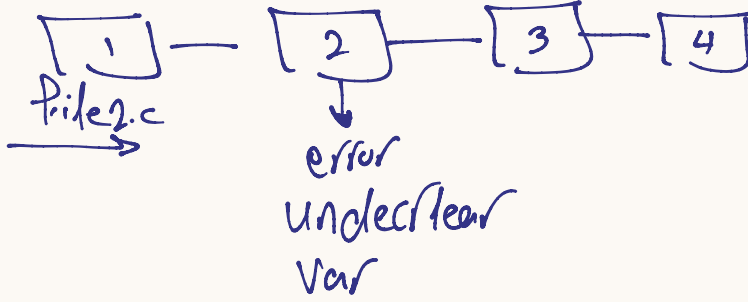


* extern
 ↳ key word → Global Function
 ↳ use

File1.c
 uint8_t global = 5;

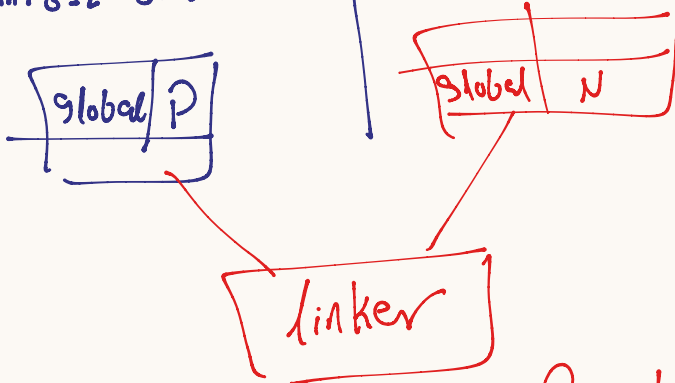
File2.c
 int main {
 printf("%d\n", global);

gcc file1.c file2.c -o out.exe

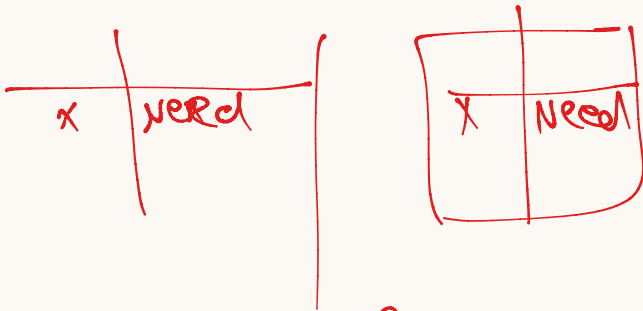


* extern → this var or this function will be need in stub table

uint8_t global = 20; extern uint8_t global;



linker will be provide error if Not found the object provide



* undefine Reference error

* static → Global Function
File1.c
 u8 speed = 120;
 static speed = 120;

File2.c
 speed = 30;

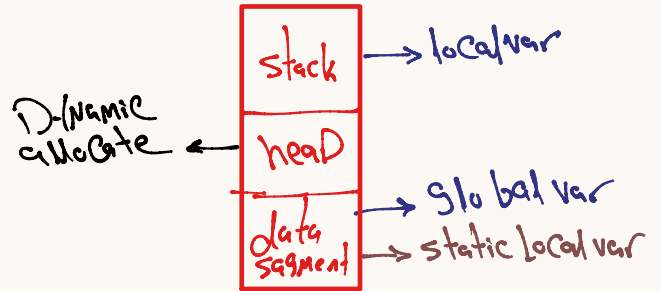
Not Add in stub table

life time

scope

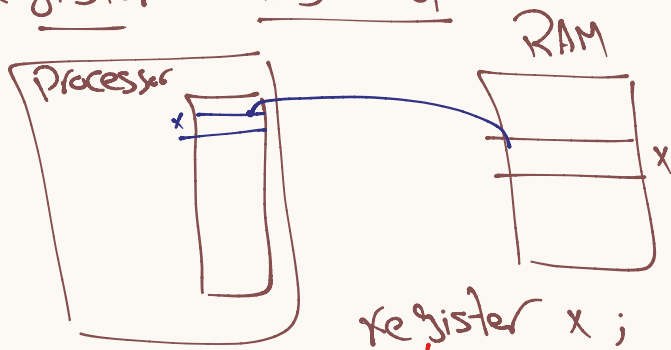
	Global	Function
life time	still Program life	still Program life
scope	change to file scope	change to file scope

* static → local var
 scope → life
 Not change still Block scope
 Change to Program life
 static local var will be store in data segment section
 RAM



static
extern
volatile
register

* register → keyword



register x;

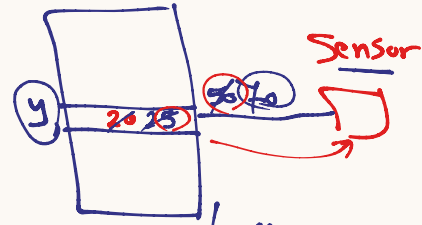
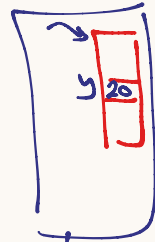
local var →

register → Request

* update in compiler

* if compiler find var used
→ will be optimized → by store this var in Processor

Processor Miss → Information



* to solve this Issue and tell compiler not store this variable inside Processor
→ will used → volatile

volatile u8 x;

* Part 2 → user define data type

Data type

Primitive	Non Primitive	User define
int float char double void	* Function * Array * Pointer	* struct * union * enum.

→ struct

* ① create new Data type ✓

* ② create object from Data type
[Declaration / definition]

* ③ Access the object

① How create new Data type b- (struct)?

→ ① struct struct name

```

{
    Data type var;
    Data type var;
};

struct Engineer
{
    uint8_t Name[20];
    uint8_t Age;
    uint16_t Salary;
};
    
```

→ ② typedef struct oldName

```

{
    Data type var;
    Data type var;
};

typedef struct {
    u8 Name [20];
    u8 Age;
    u16 degree;
} student_t;
    
```

→ ③ typedef struct

```

{
    Data type var;
    Data type var;
};

New Name;
    
```

* creation notes

→ don't take memory
→ message to compiler to define new Data type
→ can't create the same Data type in same project

for you → where write the Data type creation?

② How create object from user define data type.

→ ① struct Element

```

{
    uint8_t ID;
    uint8_t Price;
};
    
```

* struct Element obj1; → Declaration
* struct Element obj2 = { 20, 200 }; → Definition

* How to Access this obj? error

* struct not support operation
 Element_t E1;
 Element_t E2; $\Rightarrow E3 = E1 + E2;$
 Element_t E3;

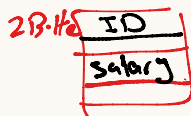
* struct support only $\rightarrow$ (Assign)
 Element_t E1;
 Element_t E2; $E1 = E2$ ✓

* to Access the variable inside object by dot Accesses.

Element_t food;
 food.ID = 30;
 food.Price = 1000;

* $\rightarrow$ typedef struct Element

{
 uint8_t ID;
 uint16_t salary;
}
Element;

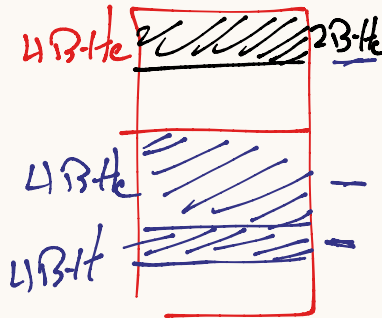


no of var *
 large data

element obj;

* typedef struct $\rightarrow$ mstruct 8 Byte

{
 uint8_t Bouns;
 uint32_t salary;
 uint8_t ID;
}
Mstruct;



total size $\rightarrow$ 12 Byte
 Actual size $\rightarrow$ 7 Byte $\rightarrow$ 5 Byte

Solution

Padding

* just write variable in order when create struct

* Reduce for Byte not used $\rightarrow$

Packing

tell compiler the size for 1 step

#pragma Pack (1)
 struct

$\rightarrow$ ② typedef struct Element

{
 uint8_t ID;
 uint8_t Price;
}
Element_t;

* Element_t obj1;
 * Element_t obj2 = {3, 1000};
 * struct Element obj3;
 * struct Element obj4 = {1, 40000};

Enum

50 $\rightarrow$ Pass
 60 $\rightarrow$ Good
 70 $\rightarrow$ very Good
 80 $\rightarrow$ excellent

$\rightarrow$ Provide Name (Descriptive, Readable)

for sum @ number

* $\rightarrow$ states $\rightarrow$ Error Code
 * $\rightarrow$ Ports

* $\rightarrow$ compiler

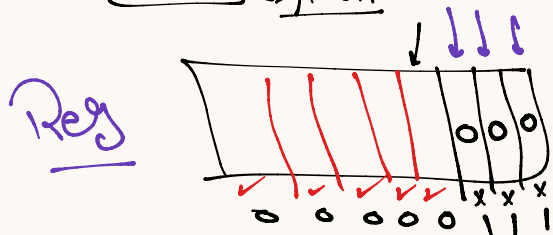
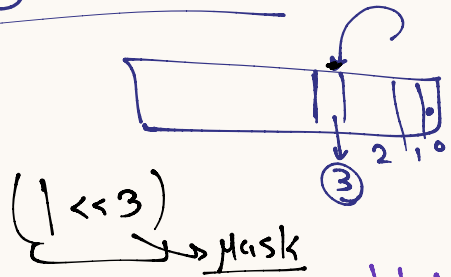
enum enumname
{
 int speed = 1;
 if (speed == 1)
 {
 Action1
 }
 else if (speed == 2)
 {
 Action2;
 }
}

typedef enum

{
 speed_low $\rightarrow$ 0
 speed_Medium $\rightarrow$ 1
 speed_High $\rightarrow$ 2
}
Speed_t;

if (speed == speed_low)
{

Masking techniques

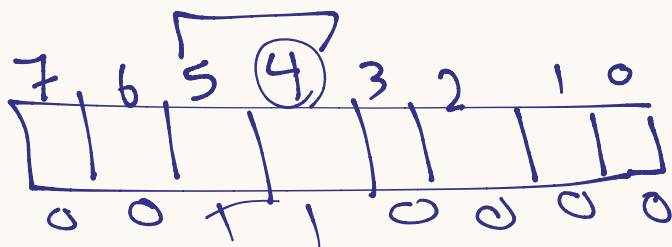
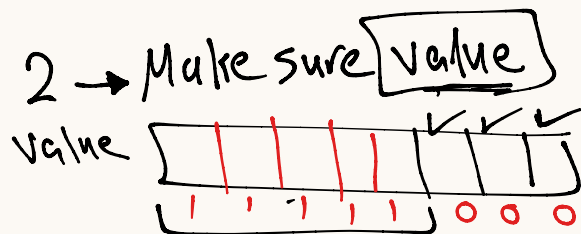


0
1
2
3
4
5
6
7
8

① → Make sure → by zero

$$= (\text{Reg} \& \sim (\text{Mask}))$$

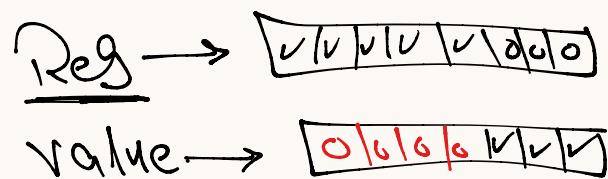
$$= (\text{Reg} \& \sim (0x07))$$



$$\text{Reg} = (\text{Reg} \& \sim \text{Mask}) | (\text{value} \ll \text{shift})$$

$$\text{Reg} = (\text{Reg} \& \sim 0x30) | ((\text{value} \ll 4) \& 0x30)$$

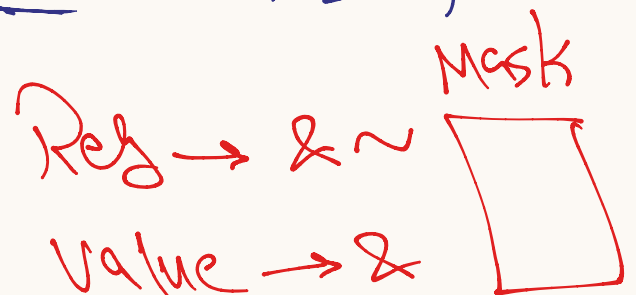
1 → set
0 → clear
^ → toggle



$$\text{Reg} = (\text{Reg} \& \sim \text{Mask}) | (\text{value})$$

0
1
2
3

Reg
Mask → 0x30 ;



000011000

* if you need control on 1 Bit.

↳ set Bit | clear Bit | toggle Bit

Read Bit

* if you need update Multi Bits

→ $Reg = [Reg \& \sim mask] | [(value \ll firstBitNo) \& mask]$

→ How get mask value

↳ write 1 in the Bits need update
otherwise zero